

A MULTI-SCALE ATTENTION-ENHANCED DEEP LEARNING ARCHITECTURE FOR ROBUST AERIAL DRONE DETECTION



RT-DETR-L · Multi-Scale CBAM · LLM Agentic Alert Layer · VisioDECT + DUT Anti-UAV

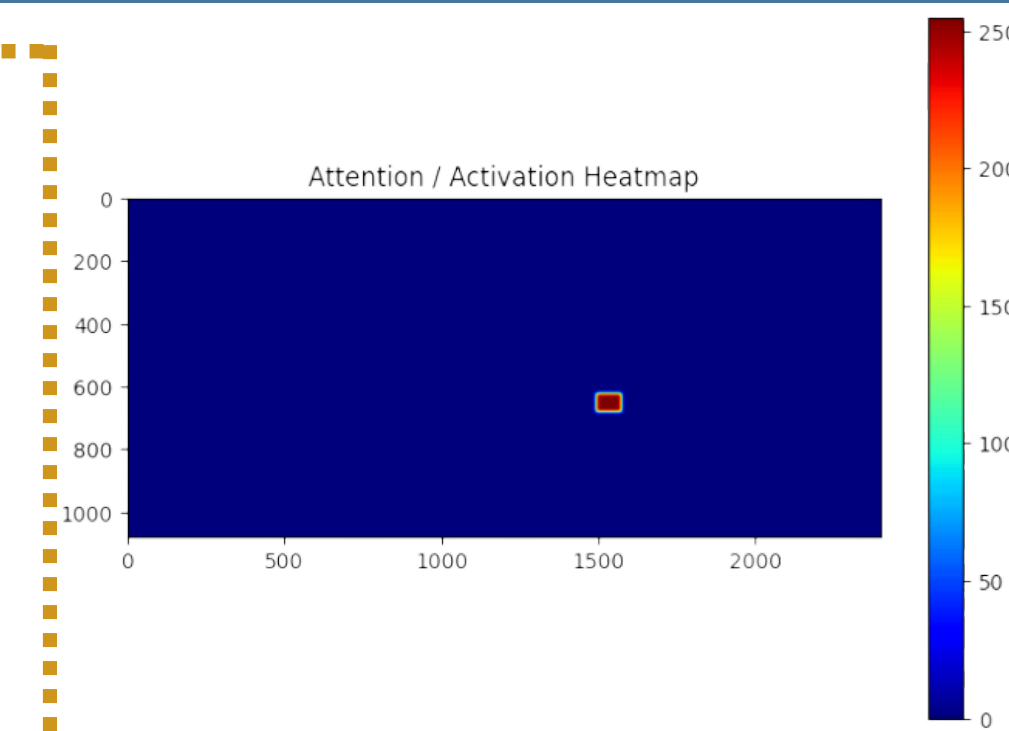
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Team Members

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Problem Statement

Detecting drones is difficult due to their small size, occlusions, and similarity to birds, while existing models lack multi-scale attention and real-time decision support



Methodology

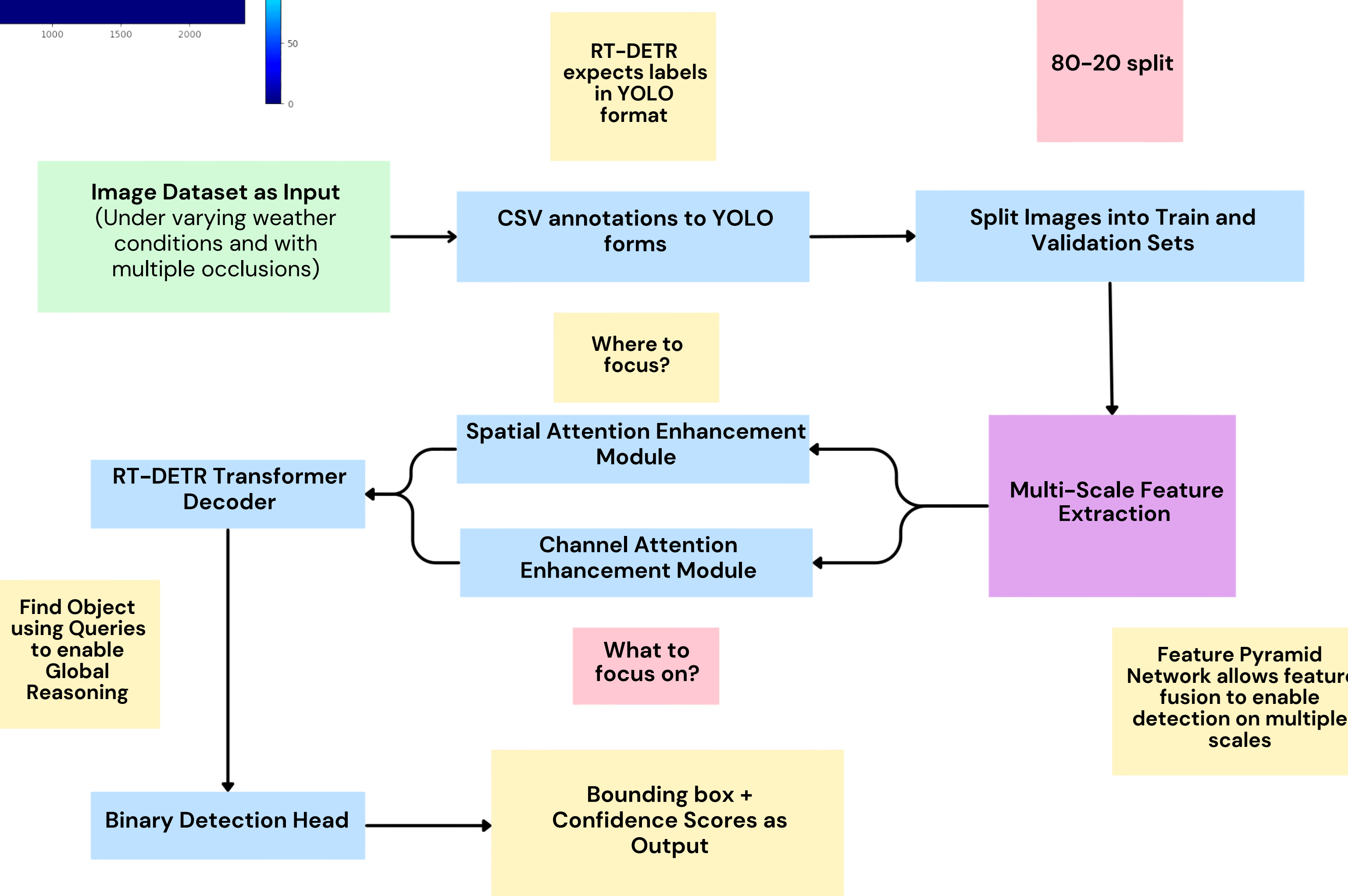
- 1 RT-DETR-L transformer for accurate drone detection
- 2 Multi-Scale CBAM (P3, P4, P5) for small and occluded objects
- 3 Occlusion-aware training using augmentation techniques
- 4 LLM layer generates intelligent, human-readable alerts

Objective

Develop a robust multi-scale drone detection system that improves occlusion handling and converts detections into actionable insights using an LLM layer.

LLM Agent Integration

Local LLM agent (llama3.2:3b via Ollama) applies 5-step ISR-style reasoning across both model outputs, session history, and spatial context – producing structured operator alerts with recommended actions. Fully offline.



Results & Performance

- mAP@0.5: 90.08%
- Precision: 90.85%
- Recall: 90.92%
- F1-Score: 90.88%
- Improved detection of small and occluded drones
- Strong performance across real-world scenarios

Applications

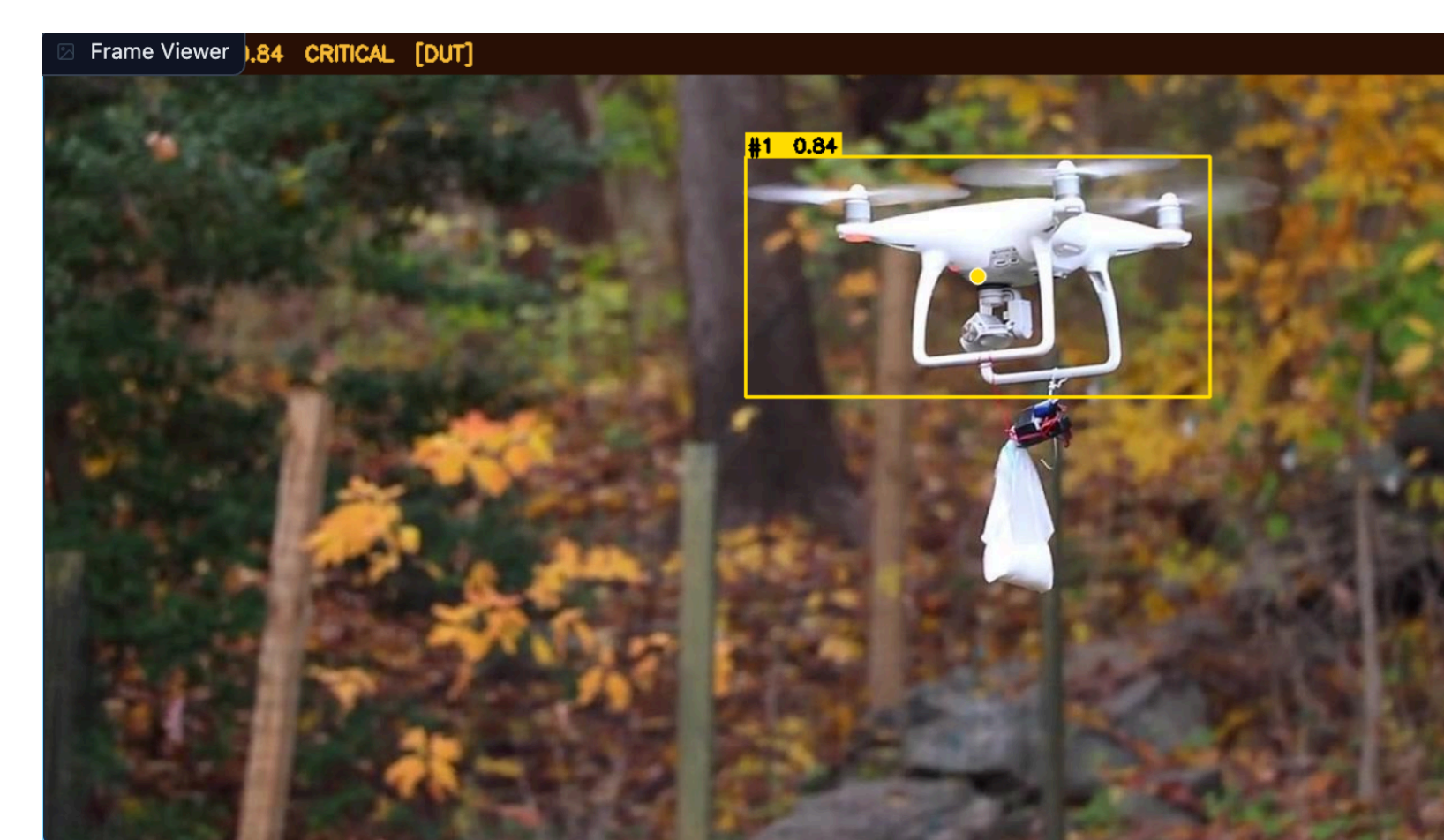
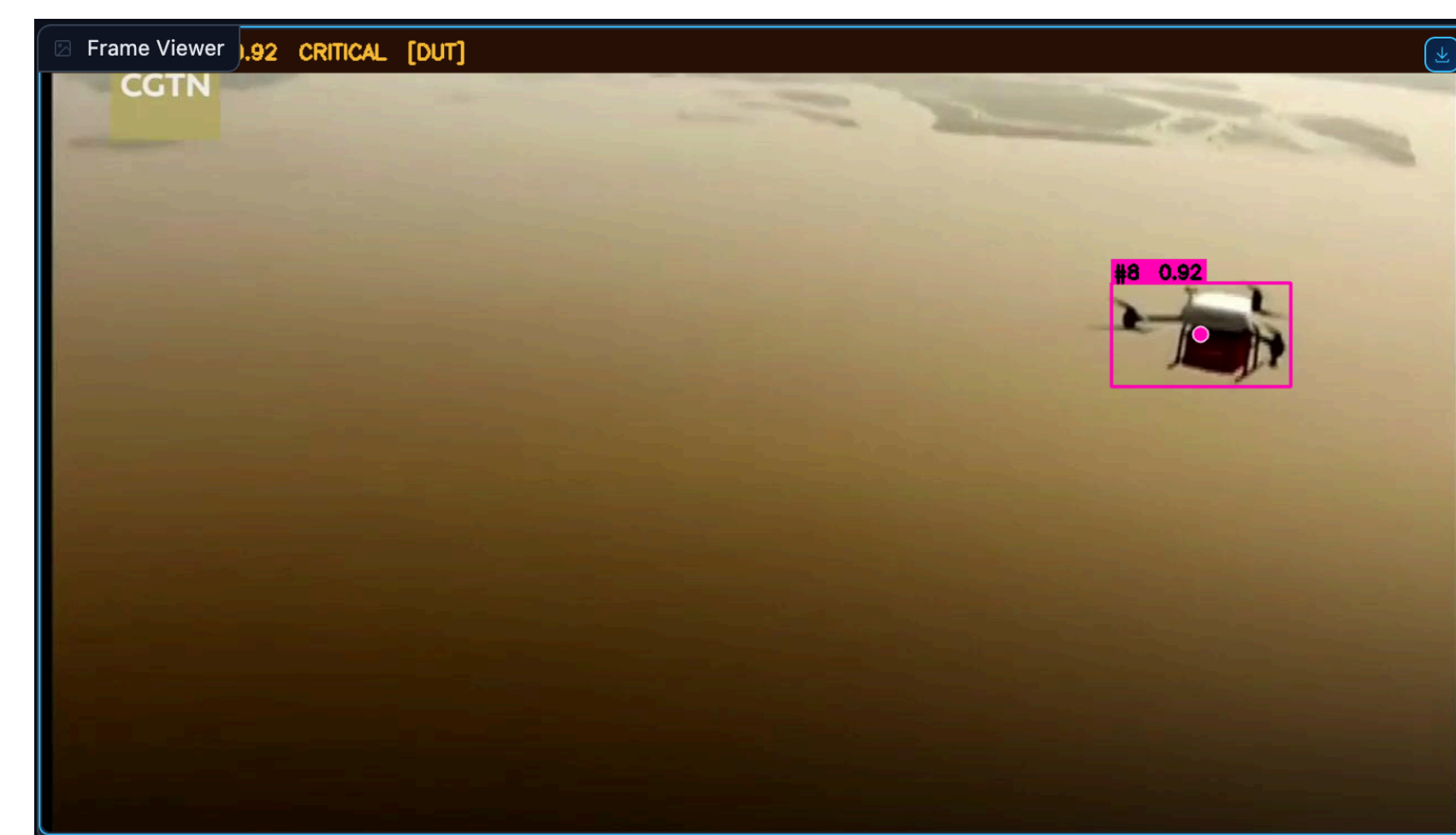
- Airspace surveillance and unauthorized drone detection
- Military monitoring and border security
- Public safety in crowded or restricted areas
- Protection of critical infrastructure (airports, power plants)

Key Conclusions

1. Multi-scale CBAM improves detection across different drone sizes
2. Strong performance even under occlusion conditions
3. LLM integration bridges detection to decision-making
4. Suitable for real-world surveillance systems

User Interface & Deployment

- Deployed as a live web application on Hugging Face Spaces
- Built using Gradio for interactive user interface
- Supports image upload, model selection, and confidence tuning
- Displays detection results with bounding boxes and confidence scores
- Generates LLM-based operator alerts with threat level and actions
- Includes frame tracking and detection logs with CSV export



Tools & Technology Stack

